

AI-Based Airport Passenger Flow Prediction System for Efficient Resource Management

Project Presentation Submitted

To

Dr. D. Y. Patil Vidyapeeth, Centre For Online Learning, Pune

Submitted By:

Name: Divya Anil Kale

ERP ID: 241114219

PRN No.: 2405020010233

Under the Guidance of

PROF. ANAND IRABATTI

Qualification : MBA ,MCA

Dr. D. Y. Patil Vidyapeeth, Centre for Online Learning, Pune

Introduction:

AI-Based Airport Passenger Flow Prediction System for Efficient Resource Management

- Airports handle large volumes of passengers every day, making efficient resource planning extremely important.
- Passenger congestion at check-in counters, security checkpoints, baggage claim areas, and boarding gates can lead to delays, overcrowding, and poor passenger experience.
- Traditional planning methods based on manual estimation are often inaccurate during sudden demand fluctuations and seasonal traffic changes.
- Artificial Intelligence and Time-Series Forecasting techniques help airports predict future passenger demand using historical travel data.
- This project uses publicly available passenger traffic data from San Francisco International Airport (SFO) obtained through the DataSF Open Data Portal.
- The study applies forecasting models such as SARIMA to identify traffic trends, seasonal patterns, and post-pandemic recovery behavior.
- Forecast results are further connected with operational planning activities such as staffing allocation, security management, check-in counter planning, and gate utilization.
- The project also demonstrates how open data and AI-based analytics can support smarter airport operations and better passenger flow management.
- The research was carried out in collaboration with NTPL Digital Services Pvt. Ltd. using the Qollabb platform for analytics learning and implementation support.

Focus Area: Passenger Forecasting | AI & ML | Airport Operations | Resource Optimization

Objective, Scope, and Purpose of Study:



Objectives

- Analyze historical passenger traffic at SFO
- Develop AI-based forecasting model
- Identify seasonal and traffic patterns
- Support airport resource planning
- Improve operational efficiency



Scope

- Uses SFO passenger traffic dataset
- Focus on monthly passenger forecasting
- Covers domestic & international traffic
- Uses SARIMA time-series model
- Supports airport operational planning

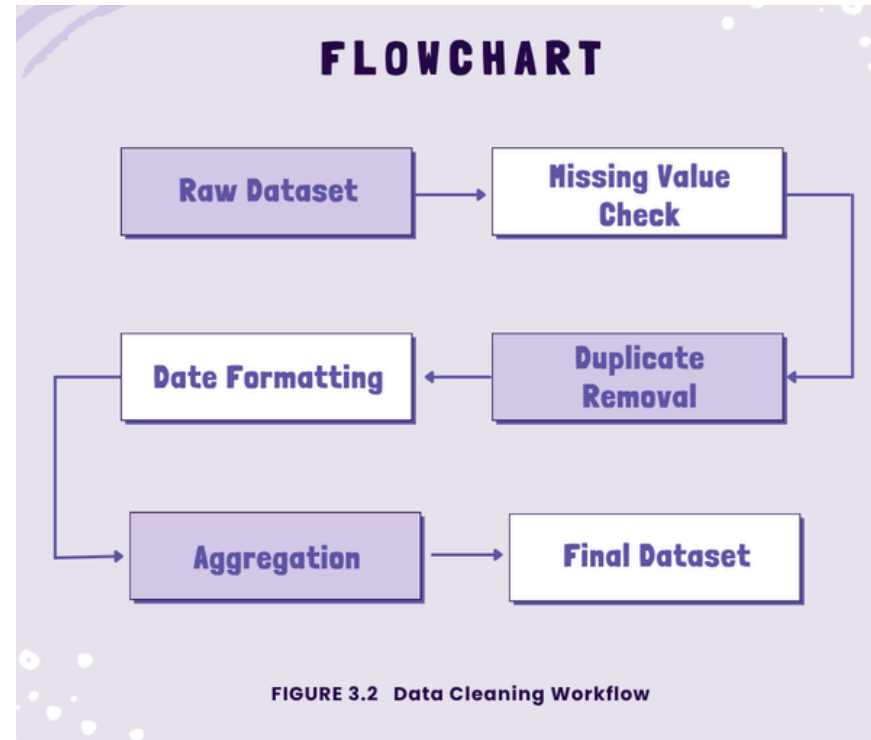


PURPOSE

- Reduce congestion and delays
- Optimize staffing and infrastructure
- Enable data-driven decisions
- Improve passenger experience
- Demonstrate AI in airport management

Literature Review

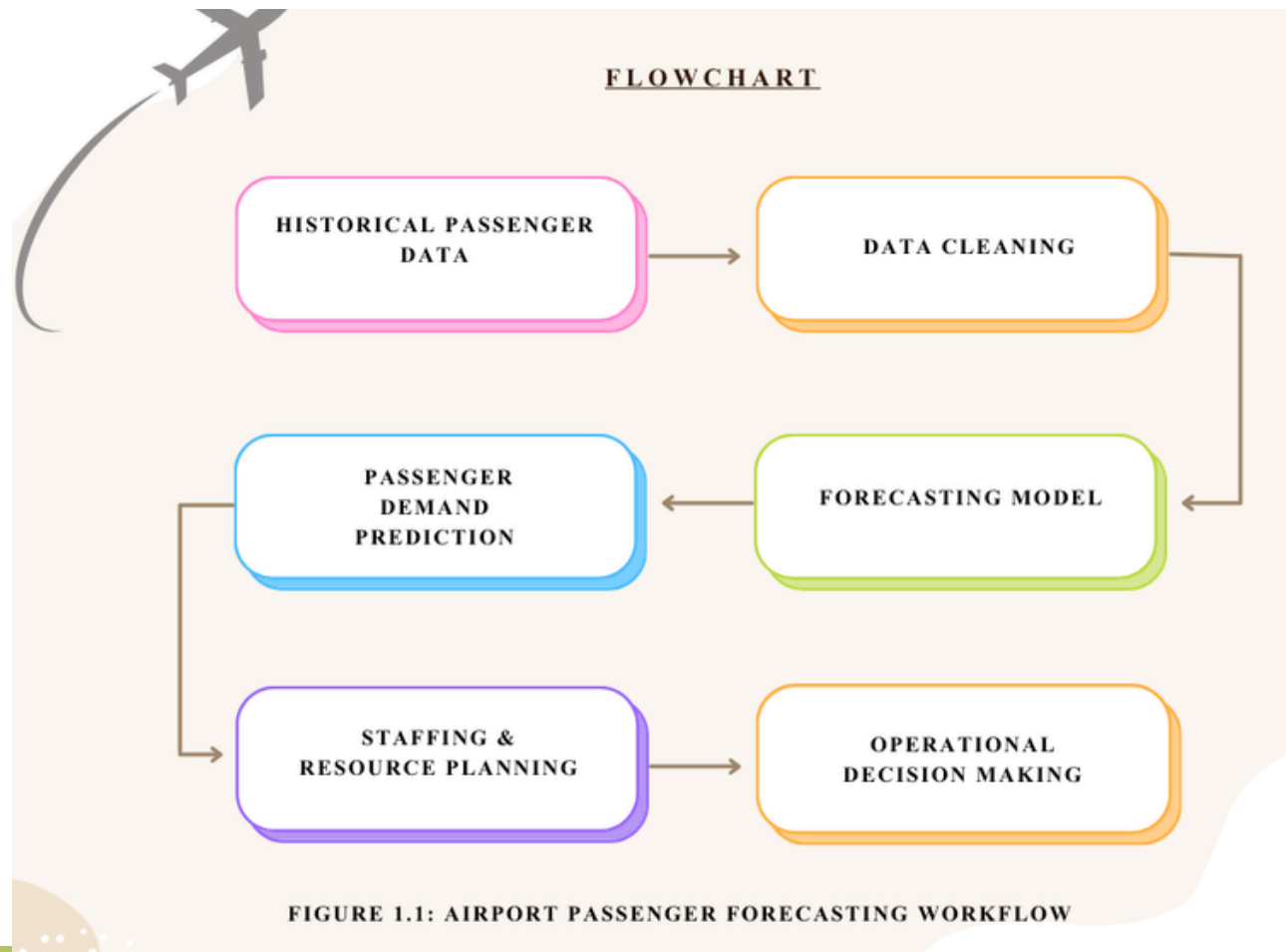
- Research Papers used for study : 12
- Published articles in Newspapers, Journal and web sites : 10
- Books : 4
- Company web sites : 6
- Study reports : 5



Key Literature Findings

- Airport passenger forecasting supports staffing, security and terminal planning.
- SARIMA and time-series models are widely used for passenger demand prediction.
- AI and machine learning improve forecasting accuracy using historical patterns.
- Passenger flow forecasting helps reduce congestion and improve service quality.
- Post-COVID studies emphasize scenario-based forecasting and uncertainty management.
- Open datasets such as SFO passenger statistics are commonly used for forecasting research.

Research Methodology

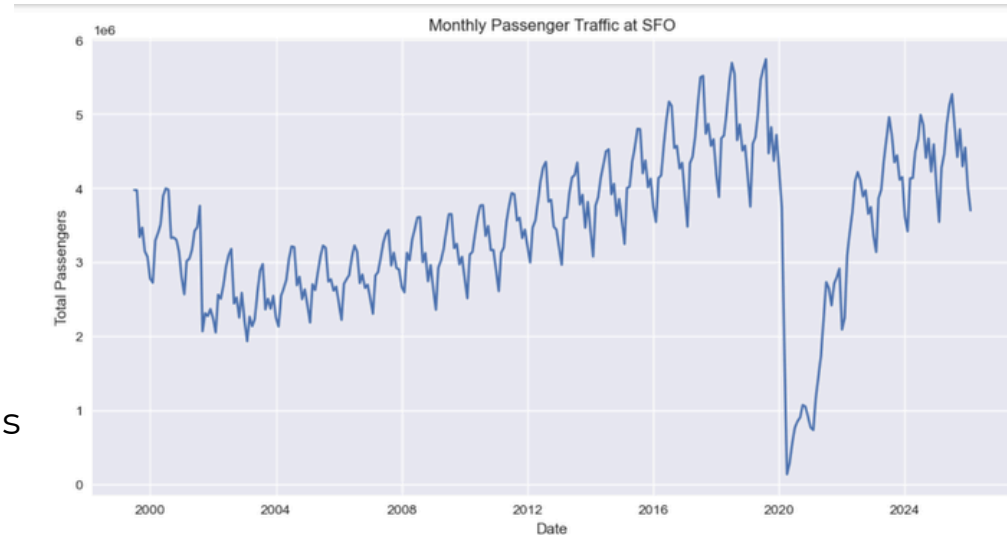


- Quantitative research approach
- Secondary data from DataSF
- Time-series forecasting using SARIMA
- Forecast validation using error metrics
- Results linked to airport resource planning

Data Collection and Analysis

Data Collection

- Data used in this study is collected from San Francisco International Airport (SFO) passenger statistics
- Source: DataSF Open Data Portal
- Dataset includes monthly passenger traffic records from 2005 onwards
- Data contains domestic and international passenger movement details
- Airline-wise and region-wise classifications are included in the dataset



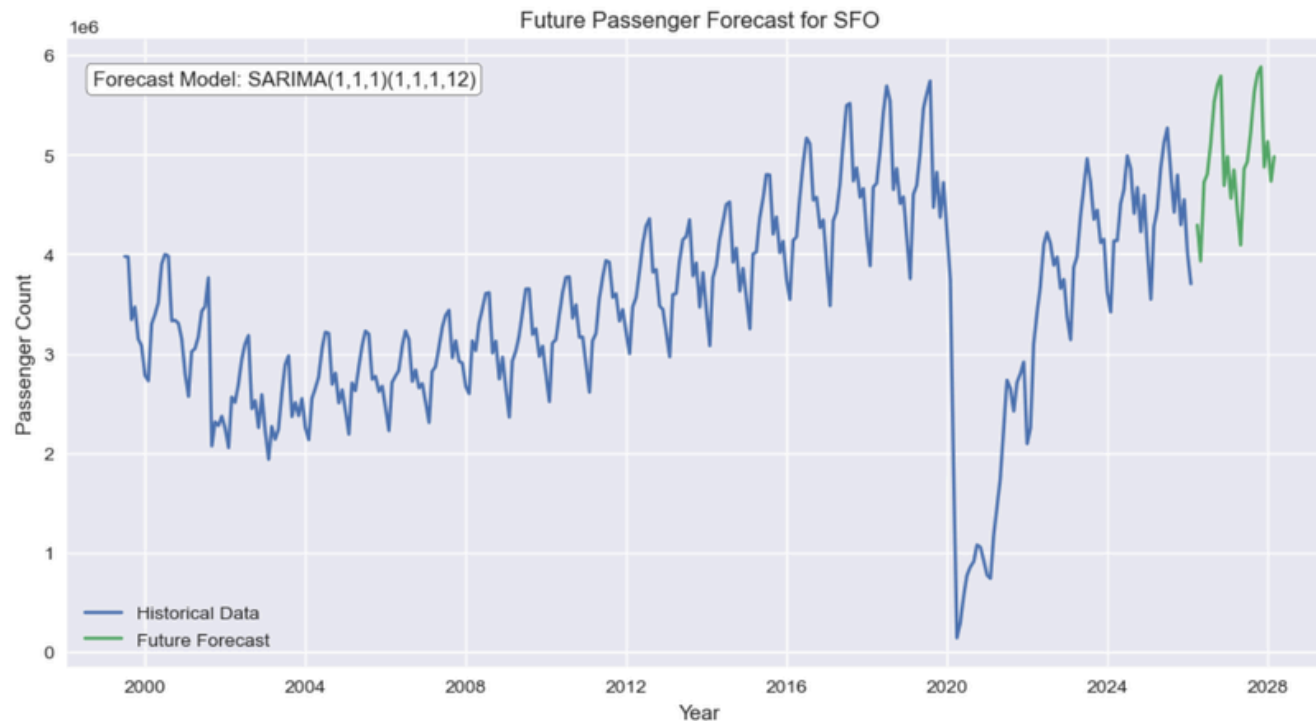
Data Preparation

- Data cleaning performed to remove missing and inconsistent values
- Unnecessary columns and duplicate records were removed
- Data was aggregated into monthly passenger totals
- Time index (year-month format) was created for forecasting

Data Analysis Approach

- Exploratory Data Analysis (EDA) performed to identify patterns
- Seasonal trends such as summer and holiday peaks were observed
- COVID-19 impact and recovery trends were analyzed
- Domestic vs international passenger trends were compared
- Time-series forecasting applied using SARIMA model

Findings



- **Key Findings from the Study**

- SFO passenger traffic shows a long-term increasing trend before COVID-19 disruption.
- Strong seasonal patterns are observed, with higher traffic during summer and holiday months.
- The COVID-19 pandemic caused a sharp decline in passenger numbers in 2020.
- Domestic passenger traffic recovered faster than international traffic after the pandemic.
- Passenger flow patterns are not linear and show repeated seasonal cycles.
- SARIMA forecasting model effectively captures trend and seasonality components.
- Forecast results show that airport demand can be reasonably predicted using historical data.
- Passenger trends are useful for resource planning such as staffing and gate management.
- Open data sources like DataSF are reliable for academic and analytical forecasting studies.
- Forecast accuracy improves when seasonal adjustments are included in the model.

AI-Based Airport Passenger Flow Prediction System

Forecasting passenger demand at San Francisco International Airport (SFO) using historical passenger traffic data and SARIMA forecasting.

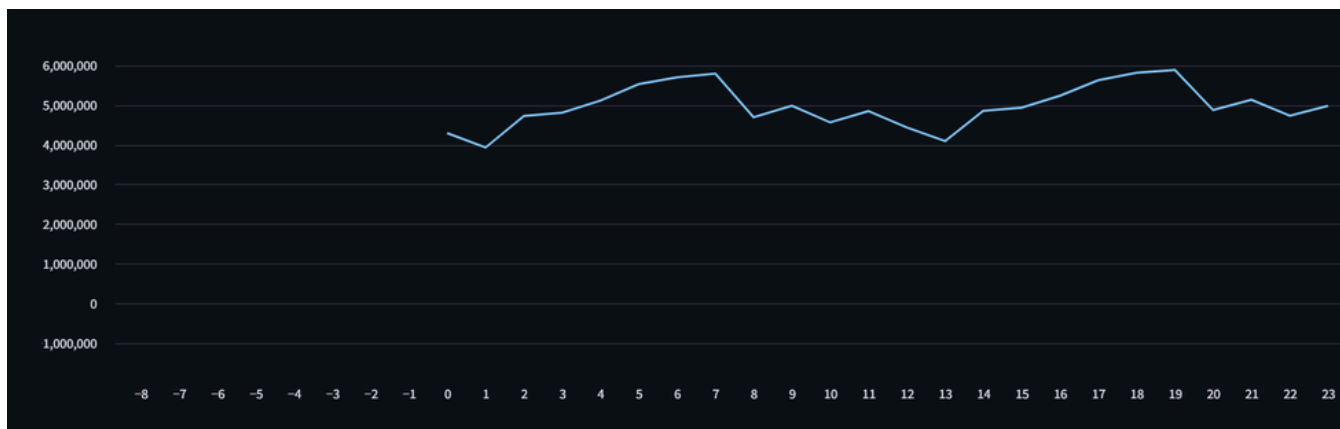
Future Passenger Forecast

Date	Predicted_Passengers
14 2027-05-31	4857500.2347
15 2027-06-30	4932631.2094
16 2027-07-31	5234492.7015
17 2027-08-31	5626707.4889
18 2027-09-30	5812474.8912
19 2027-10-31	5884109.743
20 2027-11-30	4876543.2932
21 2027-12-31	5133422.2451
22 2028-01-31	4733297.2596

-Future Passenger Forecast Table

Findings & Interpretation

- SARIMA model successfully forecasted future passenger traffic at SFO Airport.
- Predicted passenger demand ranges between approximately 4–6 million passengers per month.
- Seasonal trends are clearly visible, with higher traffic during peak travel periods.
- Forecast results support resource planning for check-in counters, security staffing, and gate allocation.
- The Streamlit dashboard provides an easy-to-understand visualization of forecast outputs for operational decision-making.



-Passenger Demand Forecast Dashboard

Conclusions

- The study demonstrates that historical passenger data from SFO can be effectively used for forecasting future airport demand using time-series models like SARIMA.
- • Clear seasonal patterns and long-term growth trends were observed, along with significant disruption during the COVID-19 period.
- • Forecasting results show that even simple statistical models can provide reliable estimates when properly cleaned and structured data is used.
- • The integration of forecasts with operational planning helps in improving staffing, resource allocation, and passenger flow management.
- • Open data sources like DataSF enhance transparency and support academic and operational research in aviation analytics.
- • However, limitations such as lack of real-time data and external influencing factors suggest scope for more advanced hybrid models in future studies.
- • Overall, the study highlights how data-driven decision-making can improve airport efficiency and support smarter infrastructure planning.

References

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Thank you

Airport Passenger Flow Management